

Does Online Mean-Reversion Beat Equal Weight?

A Reproduction and Tests of CWMR on US-Listed ETFs

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Abstract

Lahanis et al. (2025) report that mean-reversion algorithms for online portfolio selection multiply wealth hundreds to over a thousand times on NASDAQ-100 stocks over 1998–2009. We reproduce their tables exactly from their code and data, then test their variant of Confidence-Weighted Mean Reversion (CWMR) on 16 liquid US-listed ETFs over 1999–2024, deciding on each session’s close and trading at the next session’s open. We charge 0.7 basis points per dollar traded, the average half-spread we measured at 15:45 in 2016–2024; spreads at the open in earlier years were not measured and were likely wider. Traded daily, CWMR returns 12.1% a year against 8.4% for an equal-weight portfolio of the same ETFs and 8.1% for SPY, but it trades 170 times its account a year, its lead over equal weight is gone at 2.2 basis points, and the lead comes almost entirely from 1999–2008. Learning from weekly prices gives the same return with less trading (break-even 5.2 basis points), and most of that lead disappears if each trade is made one session later. Against equal weight, no return or Sharpe ratio difference is significant after correcting for the nine comparisons we report, though weekly bars’ return lead comes close (Holm-adjusted $p = 0.052$ with 126-day blocks). Against SPY, weekly bars’ return lead is significant after correction with 63- and 126-day blocks, but not with the 21-day blocks of our main specification; equal weight is the comparison that separates the algorithm from the choice of ETFs. These 1999–2024 results include the years the weekly variant was chosen on and assume costs at the open that were never measured, so they are not evidence of a tradable edge. Weekly and monthly variants passed a pre-registered test against SPY that equal weight also passes, and a monthly variant chosen on recent data failed its pre-registered test on 2025–2026.

1 Introduction

Online portfolio selection treats investing as a repeated game. Each period an algorithm holds a portfolio, observes the period’s price changes and updates its weights by a fixed rule. Cover (1991) showed that one such rule, the universal portfolio, achieves asymptotically the same growth rate as the best constant-rebalanced portfolio chosen in hindsight. A later family of algorithms bets on reversal instead. Anticor (Borodin et al., 2004), passive aggressive mean reversion (PAMR; Li et al., 2012), confidence weighted mean reversion (CWMR; Li et al., 2011, 2013), online moving average reversion (OLMAR; Li et al., 2015) and robust median reversion (RMR; Huang et al., 2013) move weight from recent winners to recent losers. The papers that introduced them report very large wealth multiples on a common set of benchmark datasets; Li and Hoi (2014) survey the family.

Lahanis et al. (2025), a publicly released working paper, implement about 25 of these algorithms and test them on NASDAQ-100 stocks; their data run from 1998 to the end of 2009, although the paper gives the sample as 1998–2010. Their best result, CWMR, multiplies wealth 1,590 times. They disclose that they ignore trading costs and that the period may not represent recent markets.

The question here is whether any version of their CWMR survives trading costs and timing on instruments a retail investor can buy today. The backtests use `btest`, the author’s backtesting software. Every result on the ETFs except the spread table and the holdout is also computed from public data by a separate backtest engine written for the replication; it ports the same algorithm code, so the agreement checks the backtests, not a second implementation of CWMR.

Related work. Trading costs have been studied in this literature from the start. Blum and Kalai (1999) extend universal portfolios to proportional costs, Li et al. (2011) report that CWMR withstands small costs on their benchmark datasets, Li et al. (2018) design an algorithm that trades reversion against costs, and Li et al. (2016) publish a toolbox for these experiments. Closest to this paper, Moon et al. (2019) test PAMR, OLMAR and cost-optimised variants on S&P 500 stocks over 2000–2017; they find that the standard benchmark datasets favour mean reversion and that the strategies may fail when explicit or implicit costs exist. At the level of assets, short-horizon reversal is documented for individual stocks at weekly (Lehmann, 1990) and monthly (Jegadeesh, 1990) horizons, while industry portfolios show momentum (Moskowitz and Grinblatt, 1999), so the stock-level evidence does not carry over directly to sector and index ETFs. Bid-ask bounce makes measured prices reverse (Roll, 1984), and part of measured short-term reversal in stocks is due to it (Kaul and Nimalendran, 1990; Conrad et al., 1997); the returns to reversal strategies can also be read as pay for supplying liquidity (Nagel, 2012). An equal-weight portfolio is a hard benchmark for optimised portfolios in general (DeMiguel et al., 2009).

Beyond Moon et al. (2019), this paper adds an exact reproduction of a recent paper, which uncovers defects in its code; a test on index and sector ETFs rather than stocks, filling at the first published price after each decision and with spreads measured from quotes; equal weight in the same funds as the benchmark; and confirmatory tests specified before they were run, with the deviations from that plan reported. Section 2 describes the data and methods. Section 3 reproduces the original paper. Section 4 tests CWMR on the ETFs over 1999–2024, Section 5 asks what costs it can bear, Section 6 examines the weekly variant and Section 7 reports the holdout test. Section 8 reports the pre-registered tests and where we departed from their plans. Section 9 discusses what the evidence supports.

2 Data and methods

2.1 Algorithm

All tests use the authors’ CWMR as written in their public repository, ported to `btest` and checked against their function to within 7.1×10^{-9} per weight on their data. Their code describes it as a simplified version of the published algorithm (Li et al., 2011, 2013). Among other differences, its step size has no confidence term, a learning-rate factor η is added and the update is not mean-centred. We keep their parameters throughout ($\epsilon = 0.89$, $\theta = 0.92$, $\eta = 0.93$) and never re-tune them.

Each period the algorithm receives a vector x of price ratios, one per asset ($x_i = 1.02$ means

asset i rose 2%). With current weights b and covariance Σ (initially the identity), it computes

$$\lambda = \eta \max\left(0, \frac{b^\top x - \epsilon}{x^\top \Sigma x}\right), \quad (1)$$

$$\mu = b - \lambda \Sigma x, \quad (2)$$

$$\Sigma^{-1} \leftarrow \Sigma^{-1} + 2\lambda\theta xx^\top, \quad (3)$$

$$b \leftarrow \Pi_\Delta(\mu), \quad (4)$$

where Π_Δ projects onto non-negative weights that sum to one. Because $\epsilon = 0.89$, the update fires unless the portfolio falls more than 11% in one period; in the weekly 1999–2024 run it fired on all but 6 of its 1,563 updates. Assets that rose most lose the most weight.

We test four schedules (Table 1).

Table 1: CWMR schedules tested.

Name	Learns from	Trades
Daily	daily ratios	every session
Monthly, daily signal	daily ratios	last session of each month
Monthly bars	month-end to month-end ratios	last session of each month
Weekly bars	week-end to week-end ratios	last session of each ISO week

2.2 Universe and data

The universe is 16 ETFs: SPY, QQQ, IWM, DIA, EFA, EEM and ten Select Sector SPDRs (XLK, XLF, XLV, XLE, XLI, XLY, XLP, XLU, XLB, XLRE). EFA and EEM hold non-US stocks and XLRE holds real estate. The ETFs launched between 1993 and 2015, so the universe grows over time: it has 11 members in January 1999 and reaches 16 in October 2015. A fund joins on its first day in the Yahoo data, which for IWM, EFA and EEM is shortly after launch. Before that it holds no weight. With fewer than two funds trading the algorithm makes no decision and the account stays in cash. Until a fund trades, the algorithm gives it the average price ratio of the funds that do and splits its weight equally among them, which gives the same portfolio return, so the algorithm runs on the whole universe from the start and keeps what it has learned when a fund joins. Until DIA starts trading in January 1998 every fund carries SPY’s price ratio, so the weights stay equal and learning in effect starts then. The universe is a set of large, liquid ETFs chosen in 2026. Funds that existed in earlier years and later closed or shrank are not in it, which favours any strategy that holds these funds, including the equal-weight benchmark.

Prices are Yahoo daily bars adjusted for dividends and splits, from 1995 so the algorithm has history to learn from before 1999. A strategy decides on a session’s close and fills at the next session’s open, the first price after its decision; in practice this would be a market-on-open order. The open is a single published price. For these ETFs’ early years we cannot check it against an opening auction, and the model fills every order at it, including on disorderly days such as 24 August 2015, when XLK traded 15% below its open within the session. Results cover 1999-01-01 to 2024-12-31; 2025 onward is a holdout (Section 7), which used Alpaca minute bars and trades at 15:45 New York time, as the author’s paper-trading system does.

2.3 Costs and benchmarks

Every dollar traded pays a flat cost in basis points (bp), plus the SEC fee on sales (the 2026 rate of \$20.60 per million dollars sold, applied to all years, although the rate varied). Turnover is dollars bought plus dollars sold per year, divided by average equity. We measured half-spreads at 15:45 from Alpaca quotes on 45 sessions in 2016–2024, five chosen at random in each year (Table 2).

Table 2: Median half-spreads at 15:45, 2016–2024.

ETF	Half-spread (bp)
SPY	0.17
QQQ, IWM, DIA	0.26–0.30
Most sector SPDRs, EFA	0.41–0.82
EEM, XLRE, XLF	1.20–1.68
Equal-weight average	0.70

We charge 0.7 bp unless stated and report results at 5, 10 and 20 bp. The model charges no commissions (Section 5 estimates what they would have cost) and no taxes (Section 9), and it ignores market impact, assuming an account small enough not to move these funds’ prices.

The benchmarks are SPY with dividends reinvested and an equal-weight portfolio (EW) of the same ETFs, reset at each month end. EW differs from the constant rebalanced portfolio (CRP) of Section 3, which resets every day. EW matters more: it holds the same funds, so a lead over it comes from the algorithm rather than from the choice of ETFs.

2.4 Statistics and procedure

We compare Sharpe ratios with the test of Jobson and Korkie (1981) with the correction of Memmel (2003), on daily returns of two strategies over the same days. The test uses their correlation, 0.71 to 0.98 across this study, so it can detect smaller differences than comparing two independent estimates. It assumes independent, normally distributed returns, and these daily returns have fat tails and clustered volatility. We therefore also report a moving-block bootstrap (Künsch, 1989) with 21-day blocks and 10,000 draws, with a 95% interval for the annual Sharpe difference, and show how its p -values change with the block length. The bootstrap is not studentised; Ledoit and Wolf (2008) propose a studentised version for this problem. We give Holm-adjusted p -values for the nine comparisons of Table 5; they do not correct for the wider search described below. The tests compare Sharpe ratios of raw returns; the tables report Sharpe ratios in excess of the 3-month Treasury bill rate.

Three tests were pre-registered: each plan was written into the research log and committed to git before the test ran, naming the strategy version, window, settings and pass criterion, and stating that the strategy would not be changed and rerun on that window (Section 8). The texts are in the replication repository with their commit hashes and times. Git commit times come from the author’s machine. For the holdout test, GitHub records the push at 10:06:06 UTC and the backtest database records the run at 10:06:32 UTC. For the 1995–2015 plan (commit d9a912d), GitHub records the push at 08:12:12 UTC, before the results commit at 08:12:49 UTC by the author’s clock. For the 1999–2015 plan (commit 97b0d4a) there is no third-party record; the order rests on commit times set by the author’s machine, public in the btest repository. The order of the work matters for what is out of sample. We first ran the original paper’s strategy families on 2016–2024 minute data, chose the monthly schedule with a daily signal, and pre-registered it for

the 2025 holdout, where it failed (Section 7). Only then did we try weekly and monthly bars on 2016–2024; weekly bars beat SPY there with calendar weeks but not at any of five other week-end offsets. Next came the 1995–2015 and 1999–2015 plans (Section 8). Everything in Sections 4 to 6 came after and is exploratory, and 2016–2024 is not out of sample for any schedule. The 2025 onward holdout had been used by two unrelated single-ETF strategies before this study and by one test in it.

3 Reproducing the original paper

The authors’ repository contains their data, 93 NASDAQ-100 stocks from 1998-01-02 to 2009-12-31. Running their functions reproduces their Tables 3 and 4 (Lahanis et al., 2025) exactly: CWMR 1,590.38 $\times$, PAMR 788.26 $\times$, Anticor 36.89 $\times$, FTRL 15.25 $\times$ and the equal-weight constant rebalanced portfolio (CRP) 12.16 $\times$.

The paper describes walk-forward validation of its parameters, but the validation windows lie inside 1998–2009 and their Tables 3 and 4 report the whole sample, so these are in-sample results. Within the sample, mean reversion keeps much of its return under frictions (Table 3).

Table 3: The authors’ data, 1998–2009. Paired Sharpe test against CRP.

	CAGR	p vs CRP
CRP	23.2%	
CWMR	85.0%	< 0.0001
CWMR, 10 bp	43.6%	0.29
CWMR, one-day delay	45.6%	0.11
PAMR, one-day delay	51.9%	0.0002
PAMR, one-day delay and 5 bp	33.9%	0.49

Four of the paper’s strategies differ from the algorithms they are named after. As implemented, OLMAR and exponential gradient never change their target weights. Both reduce to CRP, and the paper reports identical numbers for all three. CWMR and RMR are modified versions of the published methods. A fifth result depends on the data: the lead of follow the regularized leader (FTRL) over CRP disappears when the 32 tickers with incomplete data are removed. Two bad prints, PCAR in February 2000 and XRAY in July 2004, account for about 30% of CWMR’s final wealth.

4 CWMR on the ETFs, 1999–2024

Table 4 runs the four schedules of Table 1 and both benchmarks over 1999–2024 at 0.7 bp per dollar traded.

Daily, weekly-bar and monthly-bar CWMR all return more than both benchmarks; the monthly schedule with a daily signal returns less. Equal weight itself barely beats SPY over the whole period (8.37% against 8.09%).

With the 21-day blocks of Table 5, none of the Sharpe differences against equal weight is significant. Against SPY, daily and weekly bars have bootstrap p -values below 0.05 but Jobson–Korkie–Memmel p -values of 0.06 and 0.10. The bootstrap p depends on the block length. The daily return difference between each variant and equal weight is negatively autocorrelated over several weeks (for weekly bars, lag-1 -0.06 and variance ratios of 0.68 at 21 days and 0.50 at 126), so short

Table 4: 1999–2024, at 0.7 bp per dollar traded. CAGR in % a year; Sharpe ratio over T-bills; turnover per year.

	CAGR	Sharpe	Max drawdown	Turnover
Daily	12.13%	0.56	−44.7%	170×
Monthly, daily signal	7.72%	0.38	−53.1%	14×
Monthly bars	10.66%	0.49	−61.4%	14×
Weekly bars	12.10%	0.55	−50.1%	76×
Equal weight (EW)	8.37%	0.42	−54.4%	0.3×
SPY	8.09%	0.40	−55.2%	0×

Table 5: Paired Sharpe ratio tests, 1999–2024: annual difference in Sharpe ratios of raw returns, Jobson–Korkie–Mermel p , block-bootstrap p (21-day blocks), its Holm adjustment for the nine comparisons, and the 95% bootstrap interval.

	Difference	JKM p	Bootstrap p	Holm	95% interval
Daily vs EW	+0.13	0.09	0.066	0.46	[−0.01, +0.27]
Monthly, daily signal vs EW	−0.05	0.53	0.45	1	[−0.18, +0.07]
Monthly bars vs EW	+0.05	0.46	0.40	1	[−0.07, +0.18]
Weekly bars vs EW	+0.12	0.14	0.083	0.50	[−0.02, +0.27]
Daily vs SPY	+0.15	0.06	0.043	0.35	[+0.01, +0.30]
Monthly, daily signal vs SPY	−0.03	0.73	0.66	1	[−0.16, +0.10]
Monthly bars vs SPY	+0.07	0.38	0.28	1	[−0.06, +0.22]
Weekly bars vs SPY	+0.14	0.10	0.031	0.28	[+0.02, +0.28]
EW vs SPY	+0.02	0.63	0.55	1	[−0.04, +0.09]

blocks overstate its variance and longer blocks mostly give smaller p -values. With 126-day blocks, weekly bars reaches 0.048 against equal weight and 0.012 against SPY. After a Holm correction for the nine comparisons, the smallest p at any block length is 0.11 (weekly bars against SPY, 126-day blocks), and against equal weight it is 0.35.

Returns tell a stronger story than Sharpe ratios. With the same bootstrap on daily log returns and 21-day blocks, daily CWMR’s lead over equal weight is +3.4 percentage points a year (95% interval +0.6 to +6.3, $p = 0.021$), and weekly bars’ the same (+0.6 to +6.3, $p = 0.021$). Across block lengths the unadjusted p for these two leads runs from 0.006 to 0.051. After a Holm correction for the nine comparisons, the leads over equal weight stay above 0.05 at every block length, though weekly bars reaches 0.052 with 126-day blocks and 0.072 with 63-day blocks. Weekly bars’ lead over SPY is significant after correction with 63- and 126-day blocks (0.043 and 0.024; unadjusted 0.005 and 0.003), but not with the 21-day blocks of our main specification (0.085). The longer blocks fit the negative autocorrelation of the return difference, but choosing among block lengths after seeing the results is itself a selection, which Holm does not correct for. Equal weight itself leads SPY by only 0.3 points a year ($p = 0.68$), so this lead is not just the choice of ETFs; but equal weight is the comparison that isolates the algorithm, and against it the lead is not significant after correction.

5 Costs

The flat 0.7 bp is the average half-spread we measured at 15:45 in 2016–2024 (Table 2). These strategies trade at the open, and over years in which we did not measure spreads at all. Table 6 shows each schedule at higher costs, and the cost at which its return falls to equal weight’s or SPY’s.

Table 6: CAGR at higher costs per dollar traded, 1999–2024, and the break-even cost against each benchmark (bp).

	0.7 bp	5 bp	10 bp	20 bp	vs EW	vs SPY
Daily	12.13%	1.56%	−9.47%	−25.09%	2.2	2.3
Monthly, daily signal	7.72%	6.99%	6.16%	4.50%	–	–
Monthly bars	10.66%	10.01%	9.26%	7.78%	16.0	17.9
Weekly bars	12.10%	8.50%	4.46%	−3.17%	5.2	5.5

Turnover in Table 4 is dollars traded over average equity, which weights the later, larger years. Measured year by year, the daily schedule traded 230 times its account in an average year and 358 to 410 times in each year from 1999 to 2005, which is why each basis point costs it more than 170 turnover suggests.

Where each lead comes from matters for whether these costs were achievable. Daily CWMR’s lead over equal weight comes almost entirely from 1999–2008, when it returned 10.3% a year against 1.7%; over 2009–2024 it returned 13.3% against equal weight’s 12.8% and SPY’s 14.5%. Until 2001 US prices moved in sixteenths of a dollar, so a one-tick spread on a \$50 fund meant a half-spread of 6.25 bp, almost three times the 2.2 bp the daily lead can bear. Weekly bars’ lead comes mostly from 2009–2024 (18.0% against 12.8%; 3.2% against 1.7% over 1999–2008), which includes the 2016–2024 years on which it was chosen. Monthly bars can bear 16 bp, but its Sharpe ratio is not significantly better than equal weight’s ($p = 0.40$).

Estimating historical spreads from daily prices does not work at this scale. We applied the efficient estimator of Ardia et al. (2024), which its authors show improves on the earlier high–low estimators of Corwin and Schultz (2012) and Abdi and Ranaldo (2017), to each ETF’s daily open, high, low and close in each year. On 2016–2024, where half-spreads were measured at 0.17 to 1.68 bp, its yearly estimates range from −34 to +74 bp and its medians by ETF from −11 to +17 bp; for SPY it gives 11.9 bp against a measured 0.17 bp. Daily price noise swamps spreads this small, so the estimates for earlier years cannot be used.

Commissions add to this. At the 1999 average commission of the ten largest online brokers, \$15.75 a trade (Unger, 1999), the daily schedule’s 1,000 to 1,190 trades a year in 1999–2001 would have cost \$15,750 to \$18,700 a year, most of a \$20,000 account, and weekly bars’ 238 to 313 trades a fifth to a quarter of it. The backtest also buys fractional shares, which retail brokers did not offer then.

6 The weekly variant

6.1 Robustness to the day the week ends

Weekly bars end on each calendar week’s last session, usually a Friday. We reran the strategy with blocks of 5 sessions at each of the five possible starting offsets instead; holidays make these blocks drift against calendar weeks (Table 7).

Table 7: Weekly bars by where the week ends, 1999–2024, at 0.7 bp; block-bootstrap p for the Sharpe difference against equal weight.

	CAGR	Sharpe	p vs EW
Calendar weeks (Friday)	12.10%	0.55	0.083
5-session blocks, offset 0	11.01%	0.51	0.30
5-session blocks, offset 1	10.80%	0.49	0.41
5-session blocks, offset 2	13.48%	0.61	0.015
5-session blocks, offset 3	9.64%	0.44	0.86
5-session blocks, offset 4	10.13%	0.47	0.60

Every version returns more than equal weight’s 8.37%, but the Sharpe difference is significant for one of the six before any correction, and the calendar version is the second best. Which day the week ends on moves the return by almost 4 percentage points a year.

The weekly variant is also concentrated. After rebalancing it holds a median of 2 ETFs, and one ETF carries 90% or more of the account in 36% of weeks.

6.2 Trading one session late

If the weekly lead came from a reversal that unfolds over days, delaying each trade by one session should cost little. Filling at the open two sessions after the signal instead of one lowers weekly bars from 12.10% to 9.51% a year (Sharpe 0.44), and the bootstrap p against equal weight rises from 0.083 to 0.84. Most of the lead comes from the first session after the signal. Regressing each ETF’s return in excess of SPY’s on its excess return over the signal week gives +0.1 bp per 1% move from the signal’s close to the next open and -3.3 bp from that open to the open after (standard errors 0.6 and 1.2 bp, clustered by week). The reversal is between consecutive opens, so the cost of trading at the open, which we have not measured, decides whether the lead was real.

7 The holdout

The first step of the work (Section 2) was a search for a lower-turnover variant on 2016–2024 minute data, deciding at 15:30 and trading at 15:45: the paper’s 13 strategy families with daily trading, with a daily signal and monthly trades, and with monthly bars, 38 configurations in all. CWMR with a daily signal and monthly trades did best there, 18.3% a year against SPY’s 14.5%, so we pre-registered a test on 2025-01-01 onward. The pass rule was CAGR and Sharpe ratio both above SPY’s. Both came in slightly below, so it failed (Table 8).

Table 8: Pre-registered holdout, 2025-01-01 to 2026-10-02, minute data, trading at 15:45.

	Monthly CWMR	SPY
CAGR	18.28%	18.36%
Sharpe (T-bill)	0.81	0.87
Max drawdown	-20.3%	-18.7%

The paired p was 0.77, and the strategy beat SPY in 8 of 21 full months. With 438 daily returns the test cannot rule out a small edge either way. Over 1999–2024 the same variant returns 7.72% a

year, less than both benchmarks (Table 4): the lead that selected it does not hold over the longer period.

8 Pre-registered tests and deviations

Besides the holdout, two tests were pre-registered. The first asked whether the monthly variant’s lead over SPY appeared over 1995–2015, filling the years before each ETF launched with similar mutual funds and indexes. It passed, returning 11.57% a year against SPY’s 9.33%, but it decided and filled at the same close (see below). We no longer rely on it. With each ETF’s own history only it returns 6.80% against SPY’s 9.33%, mostly because it holds cash until DIA gives it a second fund; from its first trade in February 1998 to 2015 it returns 7.05% against SPY’s 6.03%.

The second tested weekly and monthly bars on 1999–2015, with CAGR and Sharpe ratio above SPY’s as the pass criterion. Both passed: weekly bars returned 9.82% (Sharpe 0.45) and monthly bars 8.90% (0.40) against SPY’s 4.90% (0.25). Equal weight passes the same criterion (6.59%, 0.33), and we already knew this basket had beaten SPY over that window when we wrote the plan, so the criterion was too easy; equal weight was the right benchmark to name.

The work departed from the plans in two ways, and one plan contained a false statement. The registered designs of both tests decided and filled at each session’s close, so each order filled at the closing price its decision had used, which no one can trade at. This look-ahead error was worth 1.7 percentage points a year to weekly bars on 1999–2015 (11.50% before the correction). We moved fills to the next session’s open and reran the 1999–2015 test with nothing else changed; a later correction to how next-open fills were booked required one further run, which left CAGR and the pass result unchanged. The holdout plan’s pipeline check judged close-to-close fills equivalent to the 15:45 fills, which missed the look-ahead error above. That plan also stated that no strategy had used the holdout window, which was wrong (Section 2).

9 Discussion

The original paper’s results are real, in the sense that its code computes them correctly from its data. On these ETFs, with fills at the next open, the daily and weekly schedules return about 3.7 percentage points a year more than equal weight at 0.7 bp (weekly bars 2.6 averaged over the five 5-session offsets). Neither lead over equal weight is statistically significant after correcting for the comparisons, though weekly bars comes close; both need costs at the open below a few basis points, the daily lead comes from years when such costs were not available, and the monthly schedule chosen for the holdout trails both benchmarks.

The weekly gain comes from a one-session reversal between consecutive opens (Section 6.2). Daily close-to-close returns of these ETFs in excess of SPY show negative next-day autocorrelation (median across the ETFs -0.036), which fits bid-ask bounce in closing prices (Roll, 1984); the strategy fills at the open, so bounce in closing prices does not explain its gain. Three explanations remain. Opening prices can bounce too: a published open may be a single trade at the bid or the ask, and a reversal between opens is then a spread the trader pays rather than a return. The reversal may be pay for supplying liquidity (Nagel, 2012), earned only by a trader whose costs at the open are low enough. And the variants came out of a search over about 100 configurations across this study (an approximate count), which inflates the best result (Harvey et al., 2016). The smallest unadjusted p in Section 4, taken over five block lengths, is 0.003, for weekly bars’ return lead over SPY with 126-day blocks ($|z| \approx 3.0$); against equal weight it is 0.006 ($|z| \approx 2.7$). Among 100 independent tests of strategies with no edge, the largest $|z|$ would exceed 3.0 about a quarter

of the time and 2.7 about half the time. The configurations are correlated, which lowers those chances, so this is only a rough guide; a deflated Sharpe ratio (Bailey and López de Prado, 2014) would make it precise.

Before calling either schedule an edge we would need:

1. spreads for these ETFs at the open over 1999–2024, measured from historical quote data, to replace the flat cost;
2. a pre-registered test against equal weight, not SPY, on data no variant has touched. The 2025 holdout has been used by one test of this study and by two earlier strategies; data from 2027 on would be clean;
3. results after commissions for a stated account size, and after tax for taxable accounts. At 76× turnover or more almost every gain is short-term, and wash-sale rules would disallow many losses.

10 Conclusion

We reproduced a publicly released set of mean-reversion results exactly, then tested the strongest algorithm on liquid ETFs over 1999–2024. Traded daily or on weekly bars it returns about 3.7 percentage points a year more than an equal-weight portfolio of the same ETFs at 0.7 bp. The daily lead needs costs at the open below about 2 bp and comes almost entirely from 1999–2008, when costs were higher; the weekly lead needs costs below about 5 bp and mostly disappears if trades are made one session later. Against equal weight, no return or Sharpe ratio difference is significant after correcting for the comparisons, though weekly bars’ return lead comes close with 126-day blocks (Holm-adjusted $p = 0.052$); against SPY, its return lead is significant with 63- and 126-day blocks but not with our main 21-day blocks. The variant selected on recent data trails both benchmarks over 1999–2024 and failed its pre-registered holdout test. These results are in-sample for the weekly variant and assume costs at the open that were never measured, so we find no evidence of a tradable edge for this CWMR variant over holding the same ETFs.

Data and code availability

Code that reproduces Section 3, Tables 3 and 4–7, Sections 4 to 6, the results of the 1999–2015 pre-registered test and the autocorrelations from public data, together with the texts of the three pre-registered plans, is at

<https://github.com/yelgabo/cwmr-etf-replication>

It uses the original authors’ public code and data (Lahanis et al., 2025), Yahoo Finance daily prices and the 3-month Treasury bill rate from FRED. The holdout test and the 2016–2024 search of Section 7 used minute bars from Alpaca, which require an account. The backtester, which computes the same tables with its own engine, and the scripts behind those sections are at

<https://github.com/yelgabo/btest>

under docs/research/olps-replication/.

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